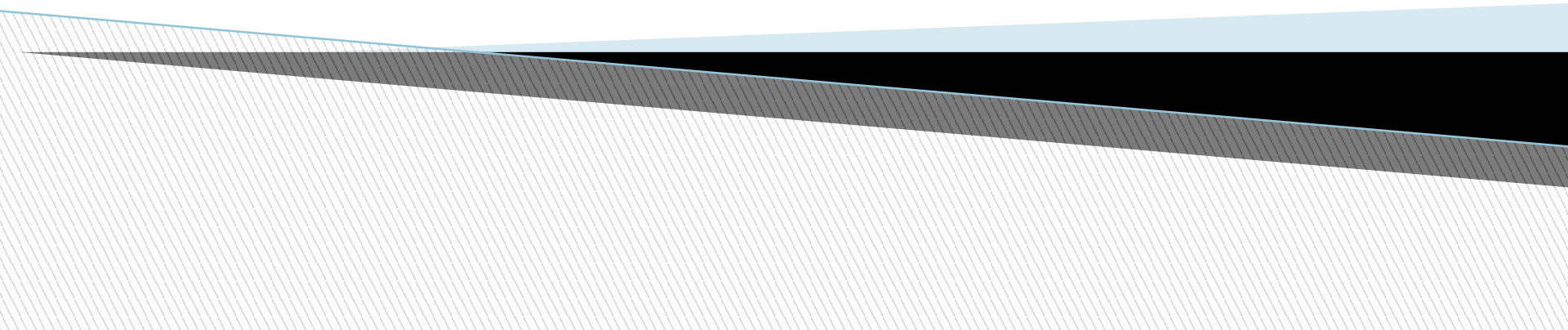
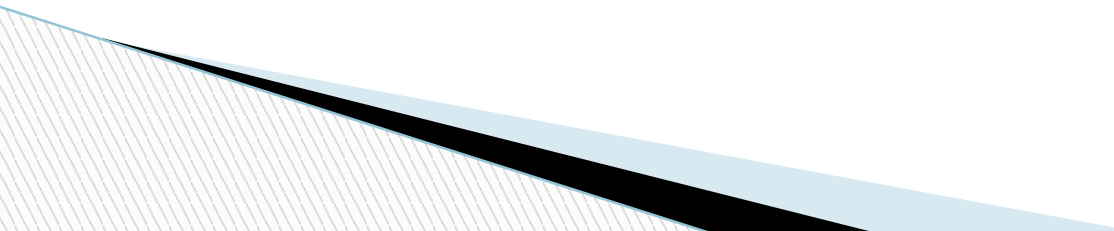


INTRUSION DETECTION SYSTEMS (IDS)

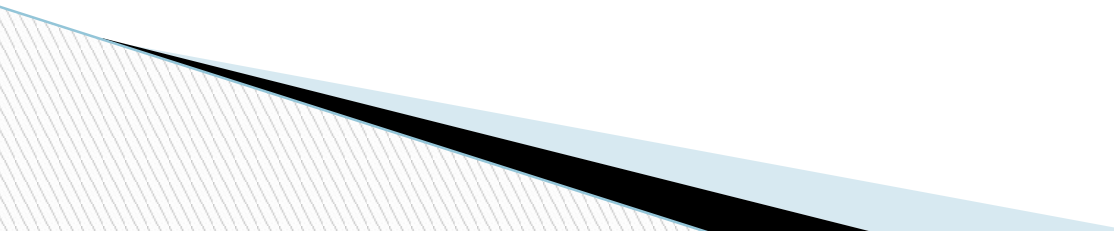
Rupali Molawade

The bottom of the slide features a decorative graphic consisting of several overlapping, wavy horizontal bands. From top to bottom, the bands are light blue, black, dark grey, and a light grey band with a fine diagonal line pattern.

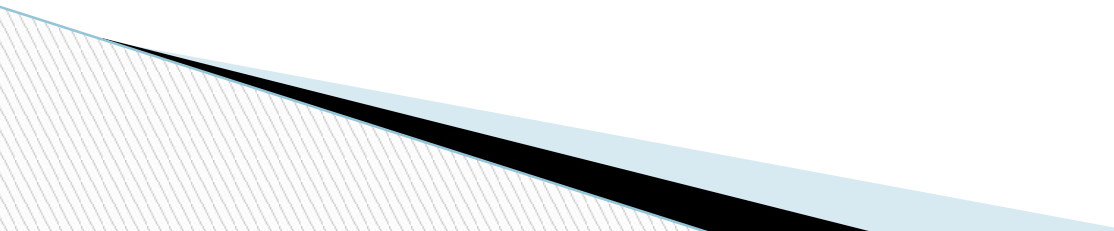
Agenda

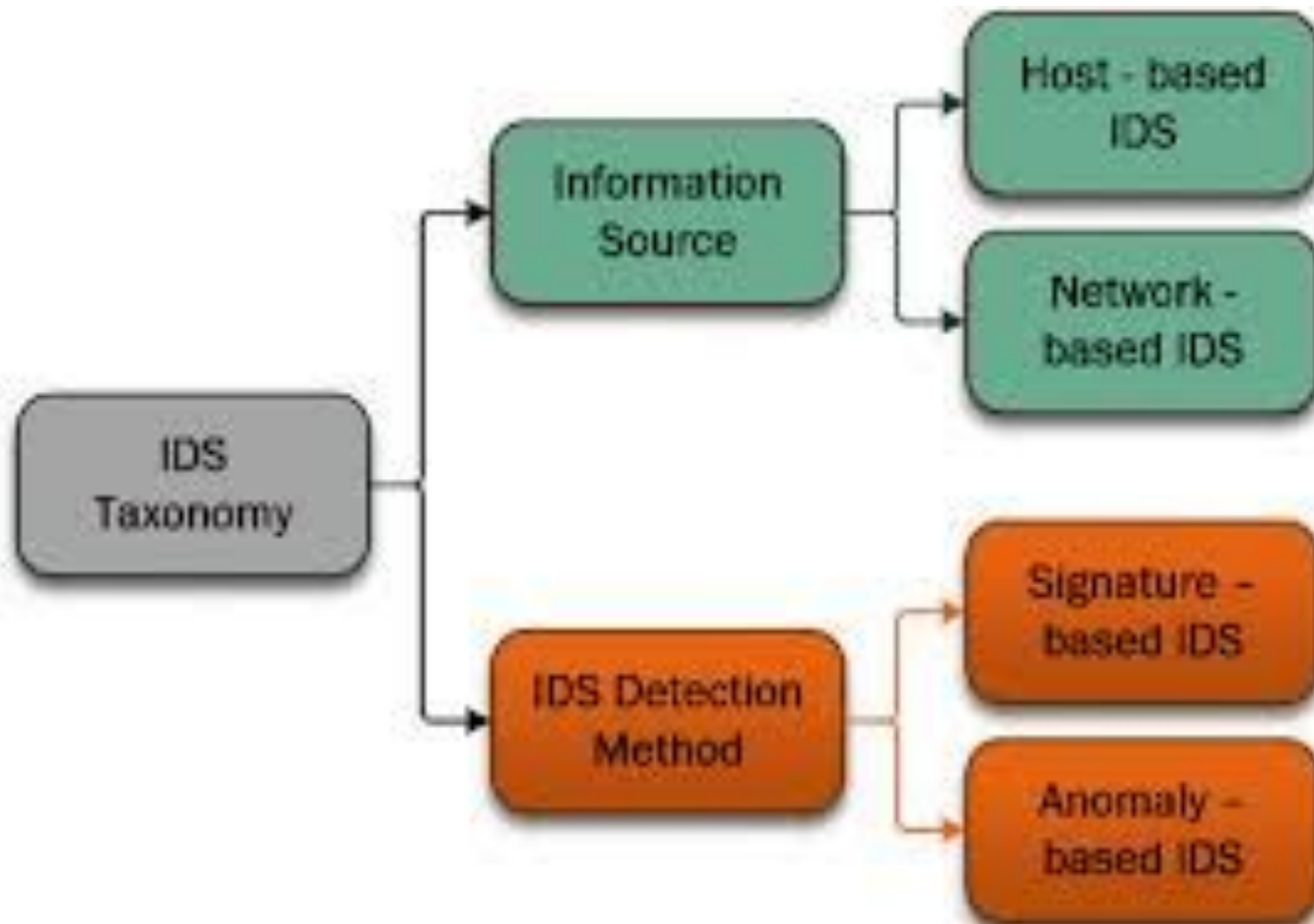
- Sources
 - What is an Intrusion Detection System
 - Types of Intrusion Detection Systems
 - How an IDS Works
 - Detection Methods
 - Issues
 - Why are IDS important
 - How does an IDS fit into your security plan?
 - Pros and Cons
 - Questions
- 

Sources

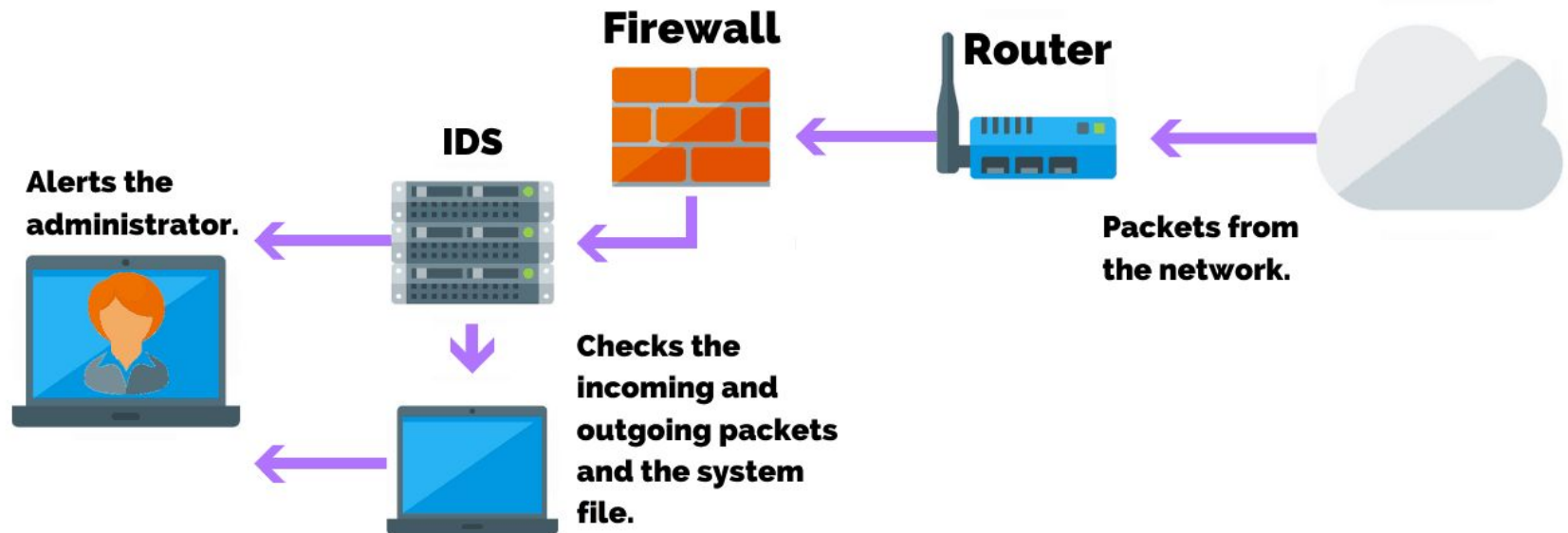
- Baker, A. R., & Esler, J. (2007). *Snort IDS and IPS Toolkit*.
 - Baumrucker, C. T., Burton, J. D., & Dentler, S. (2003). *Cisco Security Professional's Guide to Secure Intrusion Detection Systems*.
 - Endorf, C., Schultz, E., & Mellander, J. (2004). *Intrusion Detection and Prevention*.
 - Training, U. A.-I. (n.d.). Intrusion Detection Systems (IDS) and Auditing.
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What is an Intrusion Detection System?

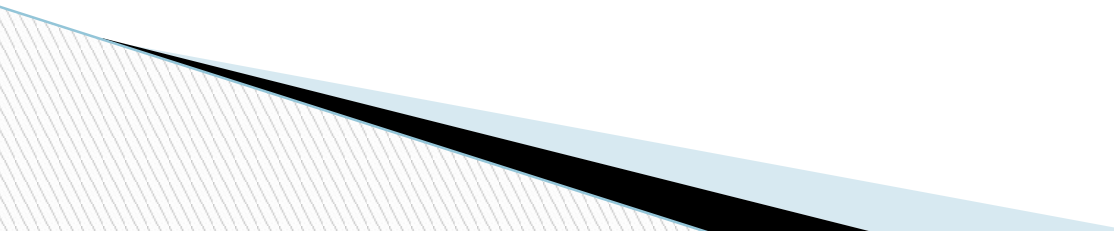
- ❑ Defined as the tools, methods, and resources to help identify, assess, and report unauthorized or unapproved network activity.
 - ❑ An IDS detects activity in traffic that may or may not be an intrusion.
 - ❑ IDSes can detect and deal with insider attacks, as well as, external attacks, and are often very useful in detecting violations of corporate security policy and other internal threats.
- 



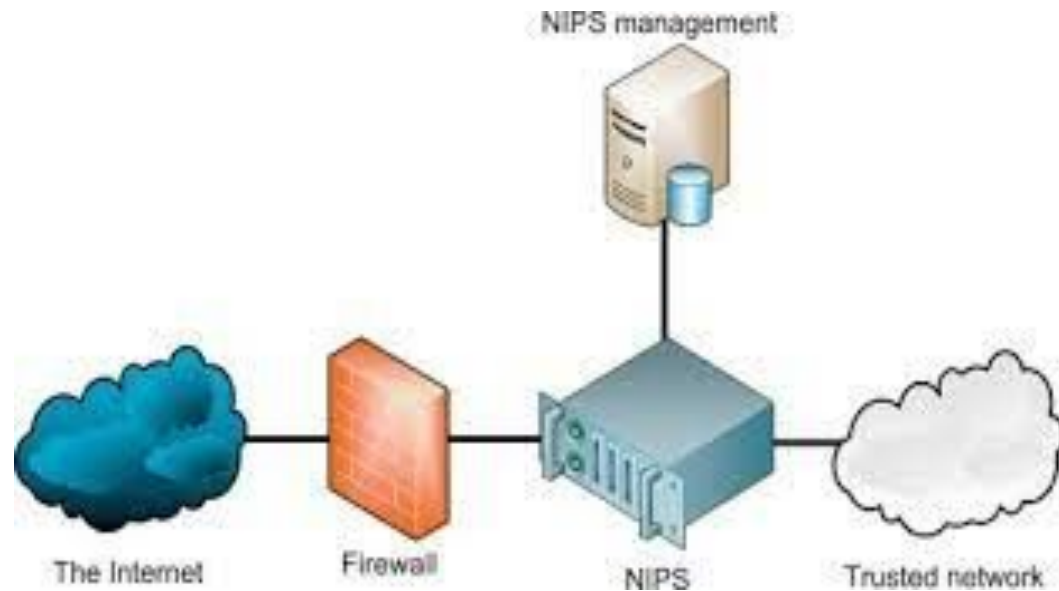
Host Intrusion Detection System (HIDS)



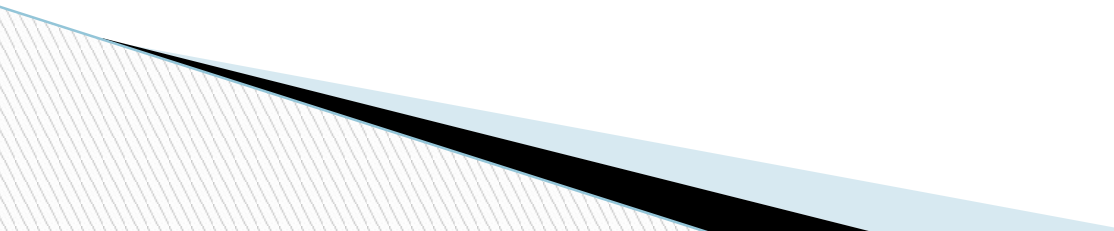
Host Based Intrusion Detection

- Are usually installed on servers and are more focused on analyzing the specific operating systems and applications, resource utilization and other system activity residing on the Host-based IDS host.
 - It will log any activities it discovers to a secure database and check to see whether the events match any malicious event record listed in the knowledge base.
 - Host-based IDS are often critical in detecting internal attacks directed towards an organization's servers such as DNS, Mail, and Web Servers.
- 

Network Based Intrusion Detection



Network Based Intrusion Detection

- ❑ Are dedicated network devices distributed within networks that monitor and inspect network traffic flowing through the device.
 - ❑ Instead of analyzing information that originates and resides on a host, Network-based IDS uses packet sniffing techniques to pull data from TCP/IP packets or other protocols that are traveling along the network.
 - ❑ Most Network-based IDS log their activities and report or alarm on questionable events.
 - ❑ Network-based IDS work best when located on the DMZ, on any subnets containing mission critical servers and just inside the firewall.
- 

Comparison

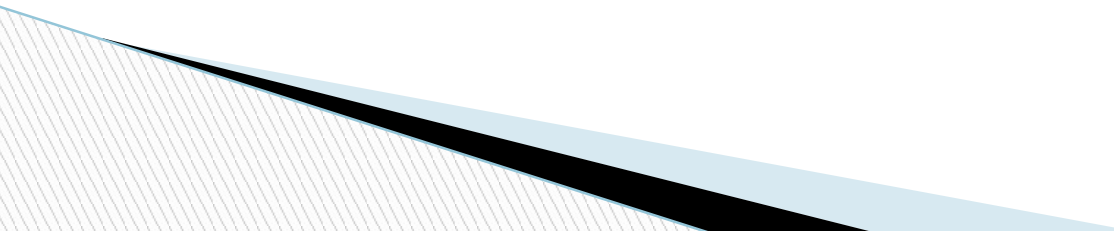
Host Based

- ❑ Narrow in scope (watches only **specific** host activities)
- ❑ More complex setup
- ❑ Better for detecting attacks from the **inside**
- ❑ **More expensive** to implement
- ❑ Detection is based on what any **single host** can record
- ❑ Does not see packet headers
- ❑ Usually only responds **after** a suspicious log entry has been made
- ❑ OS-specific
- ❑ Detects local attacks before they hit the network
- ❑ Verifies success or failure of attacks

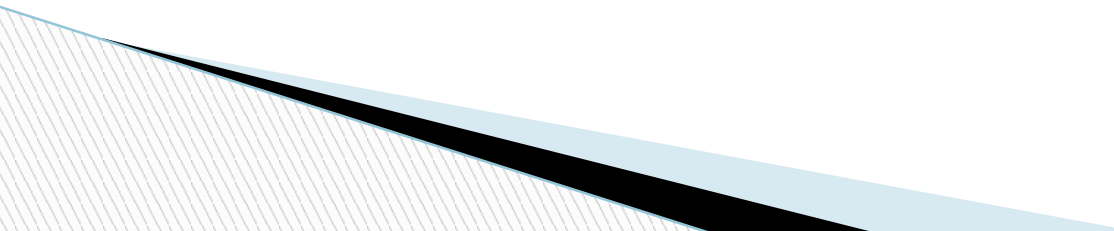
Network Based

- ❑ Broad in scope (watches **all** network activities)
- ❑ Easier setup
- ❑ Better for detecting attacks from the **outside**
- ❑ **Less expensive** to implement
- ❑ Detection is based on what can be recorded on the **entire network**
- ❑ Examines packet headers
- ❑ Near **real-time** response
- ❑ OS-independent
- ❑ Detects network attacks as payload is analyzed
- ❑ Detects unsuccessful attack attempts

Hybrid Intrusion Detection

- ❑ Are systems that combine both Host-based IDS, which monitors events occurring on the host system and Network-based IDS, which monitors network traffic, functionality on the same security platform.
 - ❑ A Hybrid IDS, can monitor system and application events and verify a file system's integrity like a Host-based IDS, but only serves to analyze network traffic destined for the device itself.
 - ❑ A Hybrid IDS is often deployed on an organization's most critical servers.
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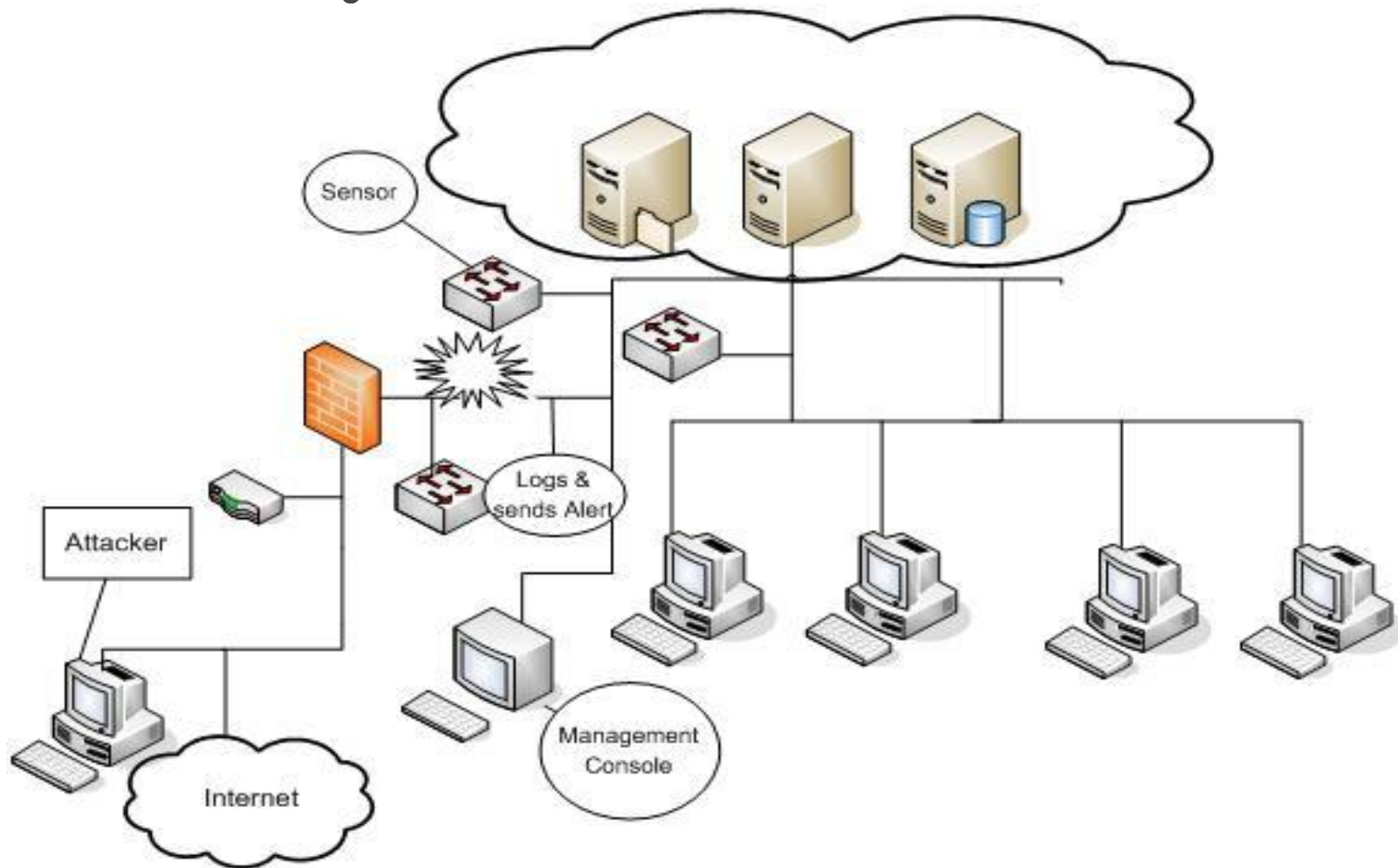
Honeypots

- ❑ Are decoy servers or systems setup to gather information regarding an attacker or intruder into networks or systems.
 - ❑ Appear to run vulnerable services and capture vital information as intruders attempt unauthorized access.
 - ❑ Provide you early warning about new attacks and exploitation trends which allow administrators to successfully configure a behavioral based profile and provide correct tuning of network sensors.
 - ❑ Can capture all keystrokes and any files that might have been used in the intrusion attempt.
- 

Passive Systems

- ❑ Detects a potential security breach
- ❑ Logs the information
- ❑ Signals an alert on the console
- ❑ Does not take any preventive measures to stop the attack

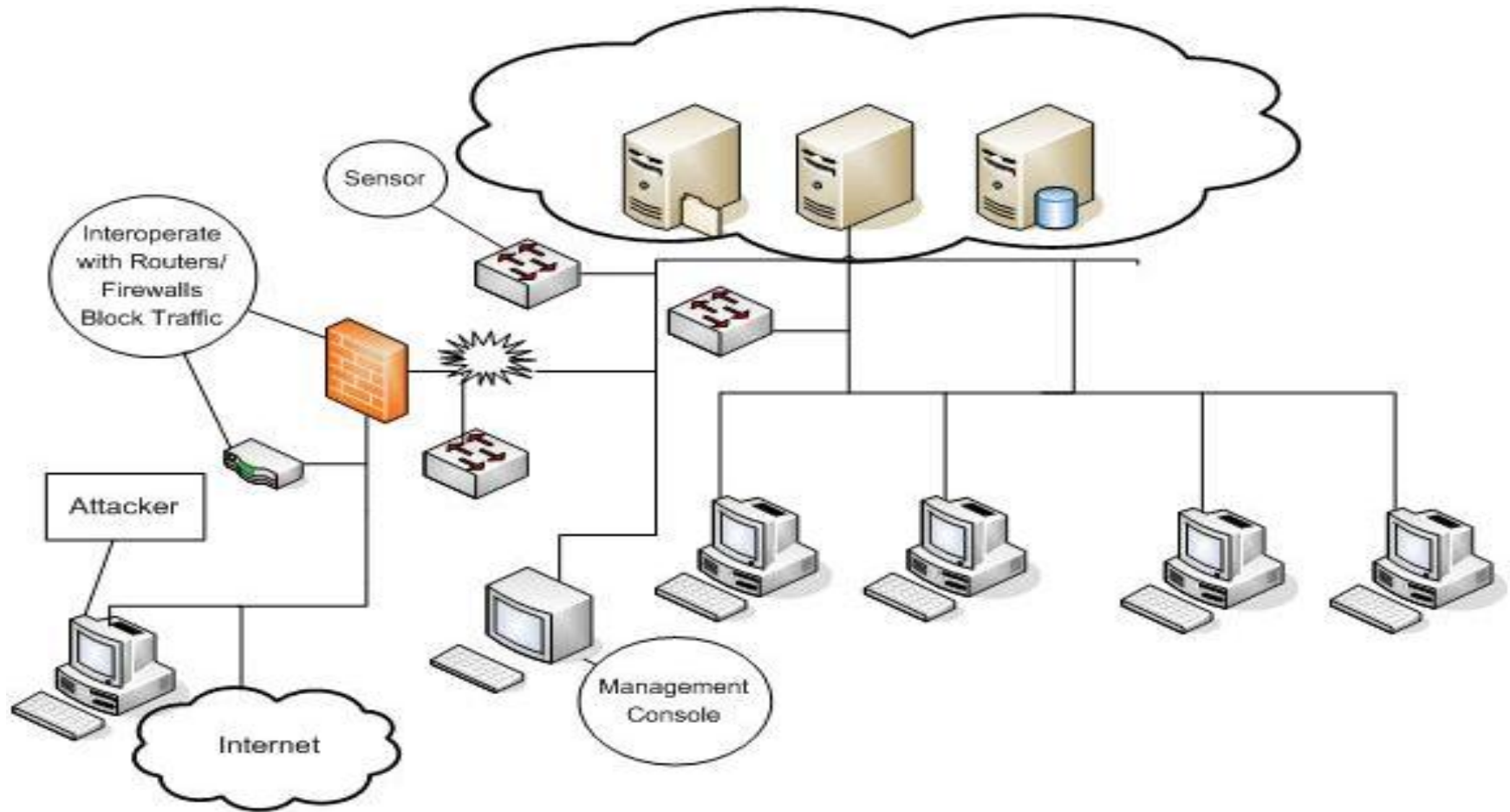
Passive Systems



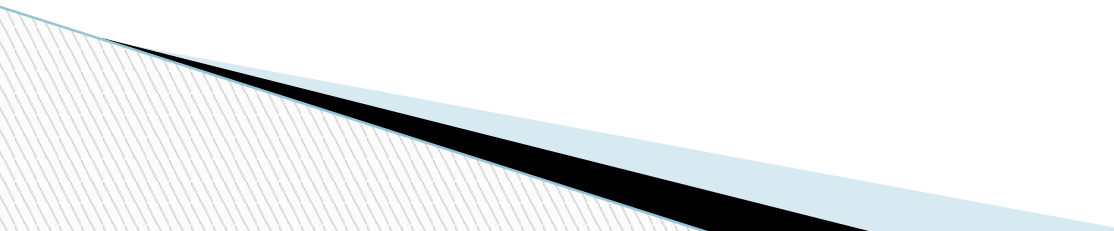
Reactive/Active Systems

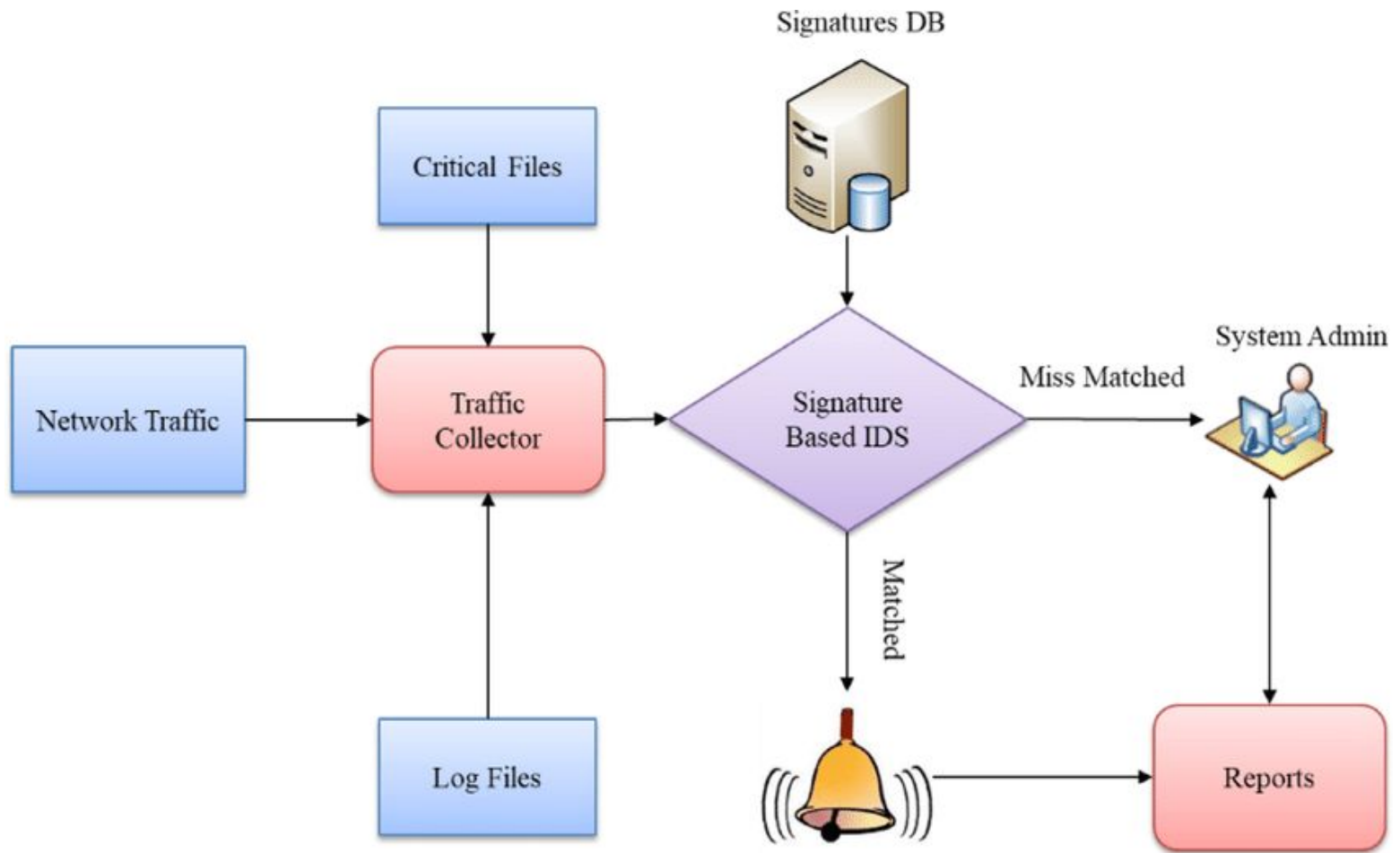
- Responds to the suspicious activity like a passive IDS by logging, alerting and recording, but offers the additional ability to take action against the offending traffic.

Reactive/Active Systems

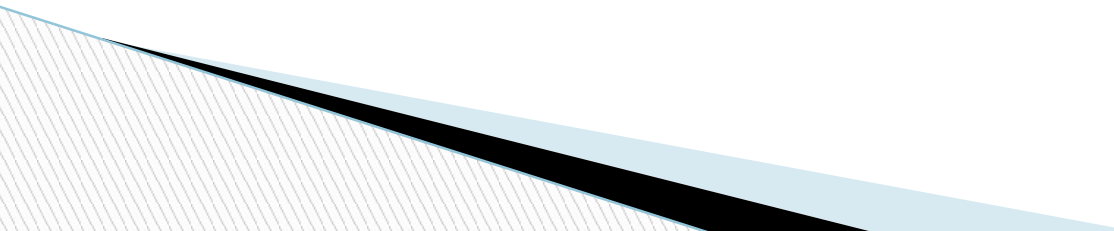


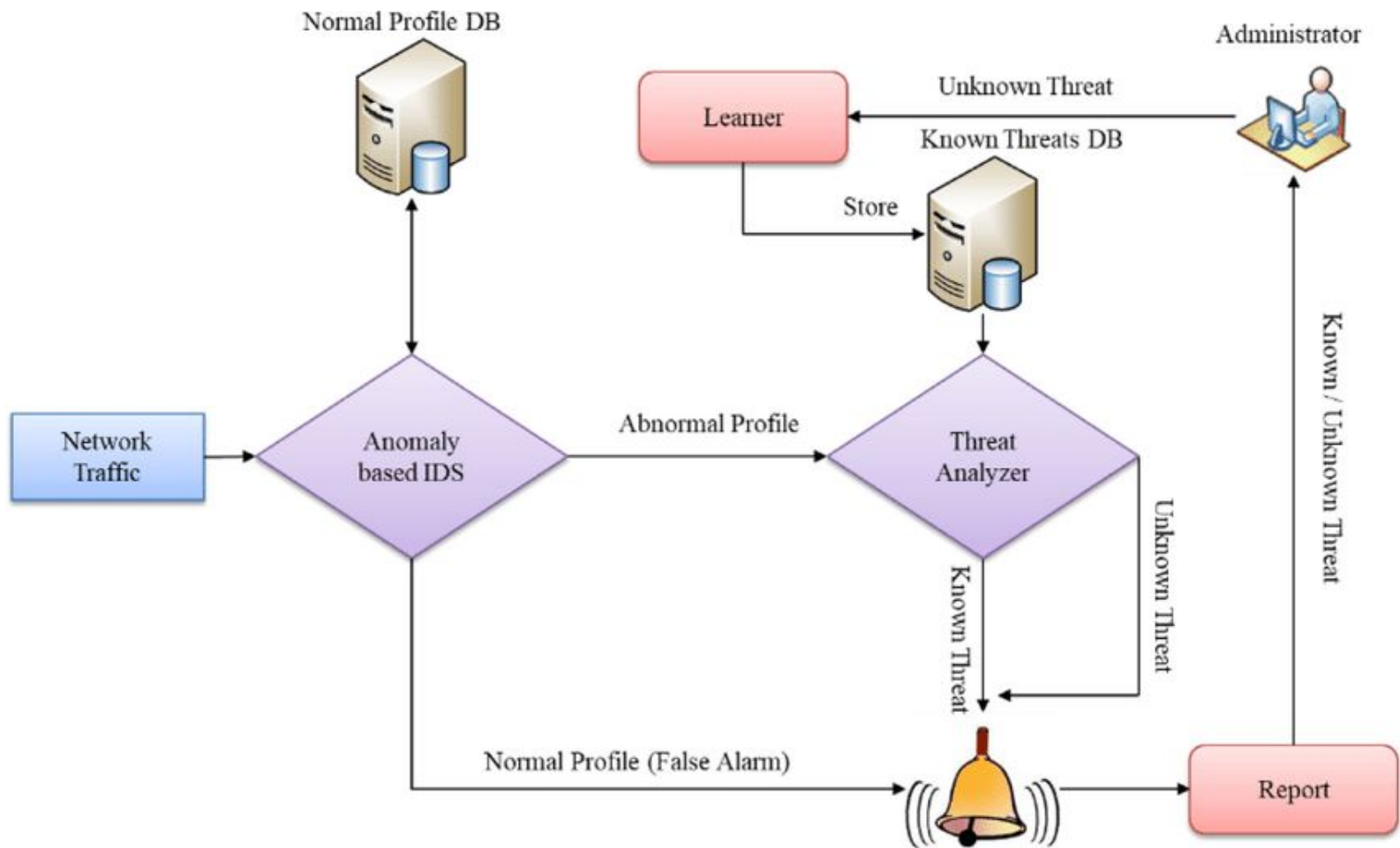
Signature Based IDS

- ❑ Monitor network or server traffic and match bytes or packet sequences against a set of predetermined attack lists or signatures.
 - ❑ Should a particular intrusion or attack session match a signature configured on the IDS, the system alerts administrators or takes other pre-configured action.
 - ❑ Signatures are easy to develop and understand if you know what network behavior you're trying to identify.
 - ❑ However, because they only detect known attacks, a signature must be created for every attack.
 - ❑ New vulnerabilities and exploits will not be detected until administrators develop new signatures.
 - ❑ Another drawback to signature-based IDS is that they are very large and it can be hard to keep up with the pace of fast moving network traffic.
- 



Anomaly Based IDS

- ❑ Use network traffic baselines to determine a “normal” state for the network and compare current traffic to that baseline.
 - ❑ Use a type of statistical calculation to determine whether current traffic deviates from “normal” traffic, which is either learned and/or specified by administrators.
 - ❑ If network anomalies occur, the IDS alerts administrators.
 - ❑ A new attack for which a signature doesn’t exist can be detected if it falls out of the “normal” traffic patterns.
 - ❑ High false alarm rates created by inaccurate profiles of “normal” network operations.
- 



Issues

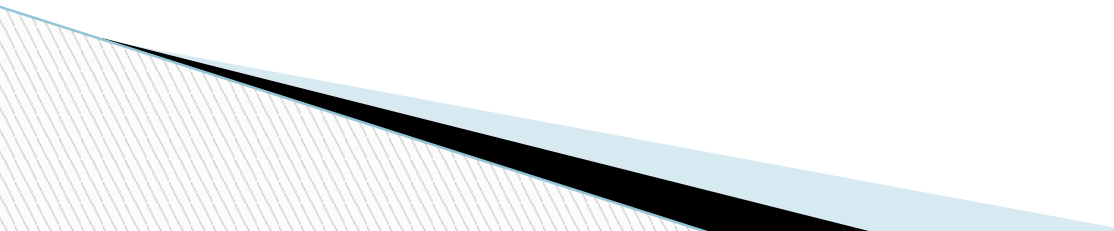
False Negatives

- ❑ When an IDS fails to detect an attack
- ❑ False negatives occur when the pattern of traffic is not identified in the signature database, such as new attack patterns.
- ❑ False negatives are deceptive because you usually have no way of knowing if and when they occurred.
- ❑ You are most likely to identify false negatives when an attack is successful and wasn't detected by the IDS.

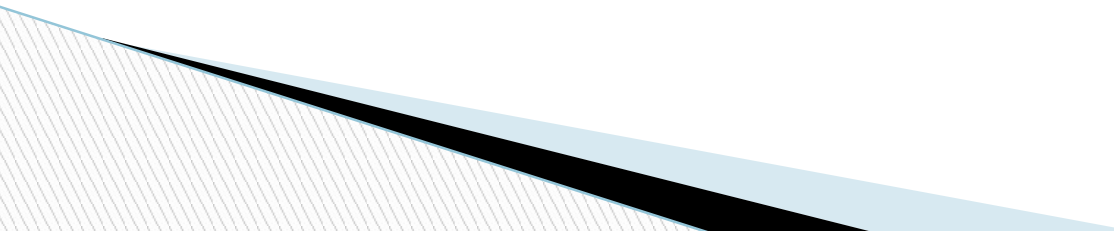
False Positives

- ❑ Described as a false alarm.
- ❑ When an IDS mistakenly reports certain “normal” network activity as malicious.
- ❑ Administrators have to fine tune the signatures or heuristics in order to prevent this type of problem.

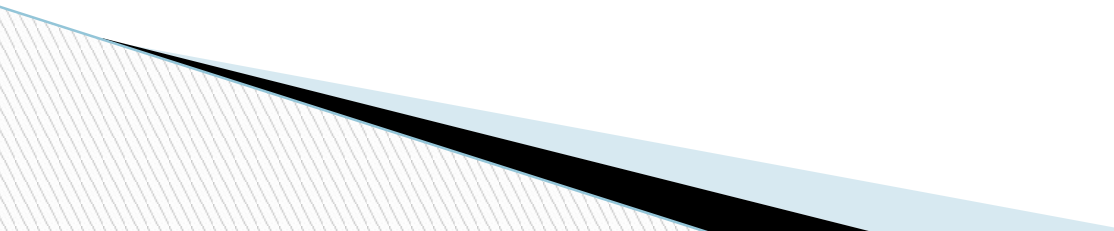
Why are IDS important?

- ❑ The ability to know when an intruder or attacker is engaged in reconnaissance or other malicious activity can mean the difference between being compromised and not being compromised.
 - ❑ An IDS can alert the administrator of a successful compromise, allowing them the opportunity to implement mitigating actions before further damage is caused
 - ❑ As Corporations and other Institutions are being legally compelled to disclose data breaches and compromises to their affected customers, this can have profound effects upon a compromised company, in the way of bad press, loss of customer trust, and the effects on their stock.
- 

How does it fit into your security plan?

- ❑ As a network security expert you should know you cannot just rely on one or a few tools to secure your network. You need to have a defense in depth mindset and layer your network defenses.
 - ❑ Through the use of inside and outside firewalls, DMZs, Routers and Switches, an IDS is a great addition to your security plan.
 - ❑ You can use them to identify vulnerabilities and weaknesses in your perimeter protection devices, such as: firewalls, switches and routers. The firewall rules and router access control lists can be verified regularly for compliance.
 - ❑ You can use IDSes to enforce security policies, such as: unauthorized Internet access, downloads of executable files, use of file sharing programs like Kazza, or Instant Messenger use.
 - ❑ IDSes are also an invaluable source of evidence. Logs from an IDS can become an important part of computer forensics and incident handling efforts.
- 

Pros

- ❑ Can detect external hackers, as well as, internal network-based attacks
 - ❑ Scales easily to provide protection for the entire network
 - ❑ Offers centralized management for correlation of distributed attacks
 - ❑ Provides defense in depth
 - ❑ Gives administrators the ability to quantify attacks
 - ❑ Provides an additional layer of protection
- 

Cons

- ❑ Generates false positives and negatives
 - ❑ Reacts to attacks rather than preventing them
 - ❑ Requires full-time monitoring and highly skilled staff dedicated to interpreting the data
 - ❑ Requires a complex incident response process
 - ❑ Cannot monitor traffic at higher network traffic rates
 - ❑ Generates an enormous amount of data to be analyzed
 - ❑ Cannot deal with encrypted network traffic
 - ❑ It is expensive
- 